SUPPLEMENTARY APPENDIX

Mapping the Political Contours of the Regulatory State: Dynamic Estimates of Agency Ideal Points

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A Agency Summary Statistics

Table A1: Agency Names and Ideal Point Estimates

Agency Id.	Est.	Department	Agency
AG-AMS	0.36	Agric.	Agricultural Marketing Service
AG-APHIS	0.25	Agric.	Animal & Plant Health Inspection Service
AG-FNS	-0.36	Agric.	Food & Nutrition Service
AG-FS	0.49	Agric.	Forest Service
AG-FSIS	0.34	Agric.	Food Safety & Inspection Service
AG-GIPSA	0.23	Agric.	Grain Inspection, Packers & Stockyards Admin.
AG-RHS	0.25	Agric.	Rural Housing Service
AG-RUS	0.63	Agric.	Rural Utilities Service
COM-BEA	-0.61	Commerce	Bureau of Economic Analysis
COM-BIS	0.75	Commerce	Bureau of Industry & Security
COM-ITA	-0.02	Commerce	International Trade Admin.
COM-NOAA	0.24	Commerce	National Oceanic & Atmospheric Admin.
COM-PTO	0.28	Commerce	Patent & Trademark Office
DOD-DARC	0.66	Defense	Defense Acquisition Regulations Council
DOD-OASHA	-1.72	Defense	Assistant Secretary for Health Affairs
DOE-EE	-0.40	Energy	Energy Efficiency & Renewable Energy
DHS-USCBP	0.07	Home. Sec.	U.S. Customs & Border Protection
DHS-USCG	0.20	Home. Sec.	U.S. Coast Guard
DHS-USCIS	-0.21	Home. Sec.	U.S. Citizenship & Immigration Services
DOI-BIA	-1.54	Interior	Bureau of Indian Affairs
DOI-BLM	0.25	Interior	Bureau of Land Management
DOI-FWS	1.74	Interior	U.S. Fish & Wildlife Service
DOI-MMS	0.81	Interior	Minerals Management Service
DOI-NPS	-0.12	Interior	National Park Service
DOJ-BOP	-0.18	Justice	Bureau of Prisons
DOJ-DEA	-1.75	Justice	Drug Enforcement Admin.
DOJ-EOIR	-2.08	Justice	Executive Office for Immigration Review
DOJ-LA	1.06	Justice	Legal Activities
DOL-EBSA	-0.81	Labor	Employee Benefits Security Admin.
DOL-ETA	0.09	Labor	Employment & Training Admin.
DOL-MSHA	-0.52	Labor	Mine Safety & Health Admin.
DOL-OSHA	-2.48	Labor	Occupational Safety & Health Admin.
DOL-PBGC	0.02	Labor	Pension Benefit Guaranty Corporation
DOT-FAA	0.35	Trans.	Federal Aviation Admin.
DOT-FHWA	0.01	Trans.	Federal Highway Admin.
DOT-FMCSA	-0.54	Trans.	Federal Motor Carrier Safety Admin.

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Agency Id.	Est.	Depart.	Agency
DOT-FRA	0.19	Trans.	Federal Railroad Admin.
DOT-FTA	0.40	Trans.	Federal Transit Admin.
DOT-NHTSA	-0.02	Trans.	National Highway Traffic Safety Admin.
DOT-OST	-0.04	Trans.	the Secretary
DOT-PHMSA	0.09	Trans.	Pipeline & Hazardous Materials Safety Admin.
ED-OESE	0.24	Education	Elementary & Secondary Education
ED-OPE	-0.20	Education	Postsecondary Education
EPA-AR	-1.63	EPA	Air & Radiation
EPA-OPPTS	-1.39	EPA	Prevention, Pesticides & Toxic Substances
EPA-SWER	-0.03	EPA	Solid Waste & Emergency Response
EPA-WATER	-0.52	EPA	Water
HHS-ACF	-1.86	Health	Admin. for Children & Families
HHS-CMS	-2.94	Health	Centers for Medicare & Medicaid Services
HHS-FDA	1.95	Health	Food & Drug Admin.
HUD-OH	0.02	Housing	Housing
HUD-PIH	-0.33	Housing	Public & Indian Housing
STATE	-0.63	State	Department of State
TR-CUSTOMS	0.02	Treasury	U.S. Customs Service
TR-DO	0.05	Treasury	Departmental Offices
TR-FINCEN	1.46	Treasury	Financial Crimes Enforcement Network
TR-OCC	1.25	Treasury	Comptroller of the Currency
TR-TTB	0.54	Treasury	Alcohol & Tobacco Tax & Trade Bureau
VA	-2.63		Department of Veterans Affairs
AID	-1.38		Agency for International Development
FAR	0.71		Federal Acquisition Regulatory Council
GSA	2.18		General Services Admin.
NARA	-0.17		National Archives & Records Admin.
NASA	-0.17		National Aeronautics & Space Admin.
NCUA	0.04		National Credit Union Admin.
OPM	-2.21		Personnel Management
RRB	-0.40		Railroad Retirement Board
SBA	2.22		Small Business Admin.
SSA	-2.67		Social Security Admin.

Table A2: Regulatory Agendas (Prop. Significant (S), Prop. Reviewed (R), and Count (N))

Agency	**Clinton**			**Bush43**			**Obama**			**Trump**		
	S	**R**	**N**	**S**	**R**	**N**	**S**	**R**	**N**	**S**	**R**	**N**
AG-AMS	0.23	0.20	35	0.12	0.12	64	0.29	0.17	42	0.22	0.22	18
AG-APHIS	0.83	0.78	46	0.68	0.65	71	0.39	0.39	66	0.15	0.15	20
AG-FNS	0.59	0.59	58	0.86	0.82	28	0.39	0.44	41	0.69	0.77	13
AG-FS	0.22	0.30	23	0.50	0.41	34	0.38	0.38	16	0.62	0.50	8
AG-FSIS	0.54	0.48	46	0.38	0.25	24	0.46	0.46	24	0.18	0.18	17
AG-GIPSA	0.35	0.35	23	0.06	0.06	17	0.36	0.36	11	1.00	1.00	1
AG-RHS	0.45	0.45	20	0.15	0.15	13	0.17	0.08	12	0.00	0.00	9
AG-RUS	0.19	0.23	31	0.17	0.13	23	0.08	0.08	12	0.00	0.00	1
AID	0.39	0.42	31	0.59	0.65	34	0.51	0.46	41	0.38	0.38	34
COM-BEA	0.00	0.00	13	0.00	0.08	25	0.00	0.00	12	0.00	0.20	5
COM-BIS	0.80	0.80	5	0.40	0.30	20	0.72	0.74	43	0.75	0.75	8
COM-ITA	0.14	0.14	7	0.20	0.20	15	0.12	0.06	17	1.00	1.00	4
COM-NOAA	0.06	0.10	398	0.13	0.13	534	0.06	0.05	528	0.07	0.07	252
COM-PTO	0.05	0.15	20	0.29	0.32	38	0.65	0.69	26	0.21	0.36	14
DHS-USCBP	0.67	0.00	3	0.57	0.39	23	0.36	0.36	14	0.27	0.18	11
DHS-USCG	0.27	0.00	15	0.31	0.19	54	0.14	0.12	56	0.11	0.11	18
DHS-USCIS	1.00	0.00	8	1.00	0.70	30	1.00	1.00	10	0.89	0.72	18
DOD-DARC	0.02	0.02	90	0.05	0.05	141	0.28	0.32	132	0.20	0.11	89
DOD-OASHA	0.04	0.39	28	0.48	0.84	31	0.21	0.18	33	0.12	0.12	8
DOE-EE	0.44	0.25	36	0.41	0.38	32	0.36	0.34	147	0.32	0.37	38
DOI-BIA	0.24	0.11	46	0.14	0.36	22	0.23	0.23	13	0.00	0.00	7
DOI-BLM	0.21	0.21	34	0.24	0.24	17	0.46	0.62	13	0.50	0.62	8
DOI-FWS	0.08	0.18	310	0.41	0.21	310	0.09	0.10	331	0.10	0.10	120
DOI-MMS	0.16	0.25	51	0.18	0.18	45	0.55	0.27	11	1.00	1.00	1
DOI-NPS	0.22	0.36	36	0.29	0.31	49	0.14	0.14	36	0.19	0.25	16
DOJ-BOP	0.12	0.03	32	0.75	0.70	20	0.53	0.47	15	0.33	0.33	3
DOJ-DEA	0.25	0.33	24	0.71	0.87	31	0.73	0.73	11	0.80	0.60	10
DOJ-EOIR	0.22	0.11	9	0.40	0.80	15	0.60	0.60	5	1.00	0.73	11
DOJ-LA	0.40	0.45	20	0.85	0.60	20	0.50	0.42	12	0.80	0.80	5
DOL-EBSA	0.59	0.65	17	0.85	0.90	20	0.96	0.86	28	1.00	0.83	12
DOL-ETA	1.00	1.00	9	1.00	0.89	18	1.00	0.92	12	1.00	1.00	8
DOL-MSHA	0.58	0.42	12	0.95	0.84	19	1.00	1.00	8	0.67	0.67	3
DOL-OSHA	0.77	0.27	22	0.72	0.83	18	0.43	0.30	23	0.89	0.89	9
DOL-PBGC	0.31	0.19	16	0.71	0.64	14	0.31	0.38	13	0.00	0.00	4
DOT-FAA	0.42	0.30	135	0.31	0.24	104	0.27	0.28	75	0.30	0.26	23
DOT-FHWA	0.31	0.23	78	0.29	0.29	38	0.42	0.39	31	0.42	0.25	12
DOT-FMCSA	0.61	0.14	36	0.63	0.63	19	0.46	0.39	28	0.20	0.24	25

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Table A2 – continued from previous page

Agency	Clinton			Bush43			Obama			Trump		
	S	**R**	**N**	**S**	**R**	**N**	**S**	**R**	**N**	**S**	**R**	**N**
DOT-FRA	0.65	0.48	23	0.35	0.26	23	0.34	0.32	38	0.25	0.25	16
DOT-FTA	0.39	0.17	18	0.38	0.19	16	0.50	0.45	20	0.33	0.00	3
DOT-NHTSA	0.11	0.11	140	0.23	0.23	101	0.30	0.26	69	0.27	0.33	15
DOT-OST	0.25	0.22	55	0.29	0.32	41	0.38	0.42	24	0.42	0.42	12
DOT-PHMSA	0.22	0.16	63	0.28	0.24	50	0.33	0.27	52	0.50	0.50	10
ED-OESE	0.80	0.60	5	0.89	0.89	9	0.73	0.82	22	0.00	0.00	4
ED-OPE	0.68	0.46	28	1.00	1.00	5	0.95	0.95	22	1.00	0.82	11
EPA-AR	0.36	0.35	291	0.37	0.51	281	0.52	0.51	263	0.38	0.38	123
EPA-OPPTS	0.59	0.45	44	0.33	0.38	21	0.56	0.54	52	0.50	0.61	28
EPA-SWER	0.65	0.57	60	0.77	0.69	35	0.48	0.48	31	0.88	0.76	17
EPA-WATER	0.41	0.42	80	0.36	0.47	47	0.49	0.49	41	0.71	0.65	17
FAR	0.17	0.24	181	0.12	0.11	105	0.62	0.59	95	0.32	0.37	41
GSA	0.32	0.35	62	0.12	0.08	48	0.17	0.12	41	0.67	0.53	15
HHS-ACF	0.55	0.71	42	0.83	1.00	18	0.93	0.86	14	0.86	1.00	7
HHS-CMS	0.72	0.83	103	0.90	0.96	174	0.98	0.92	196	0.98	0.94	90
HHS-FDA	0.62	0.48	121	0.73	0.50	62	0.62	0.50	68	0.50	0.53	40
HUD-OH	0.55	0.60	42	0.82	0.82	56	0.44	0.46	50	0.62	0.62	8
HUD-PIH	0.95	0.85	39	0.87	0.83	30	0.71	0.59	17	1.00	1.00	5
NARA	0.39	0.39	18	0.19	0.26	31	0.62	0.56	16	0.33	0.33	3
NASA	0.11	0.11	9	0.06	0.06	17	0.36	0.18	33	0.33	0.33	6
NCUA	0.00	0.00	31	0.01	0.00	72	0.05	0.00	65	0.00	0.00	38
OPM	0.14	0.54	103	0.56	0.87	104	0.53	0.61	106	0.68	0.45	38
RRB	0.10	0.10	31	0.38	0.38	13	0.20	0.00	5	0.00	0.00	2
SBA	0.63	0.52	75	0.78	0.75	51	0.61	0.65	57	0.38	0.27	26
SSA	0.44	0.32	57	0.78	0.80	80	0.89	0.68	38	0.80	0.70	10
STATE	0.11	0.33	9	0.41	0.51	37	0.78	0.69	58	0.86	0.86	21
TR-CUSTOMS	0.02	0.00	66	0.00	0.00	22	0.28	0.22	18	0.71	0.57	7
TR-DO	0.17	0.00	12	0.28	0.25	40	0.57	0.57	21	0.40	0.10	10
TR-FINCEN	0.00	0.00	11	0.06	0.03	32	0.63	0.59	27	0.75	0.25	4
TR-OCC	0.02	0.05	43	0.28	0.08	36	0.26	0.07	43	0.18	0.00	39
TR-TTB	0.00	0.00	7	0.05	0.02	55	0.05	0.05	20	0.50	0.50	2
VA	0.23	0.60	114	0.58	0.90	112	0.41	0.30	127	0.34	0.39	67

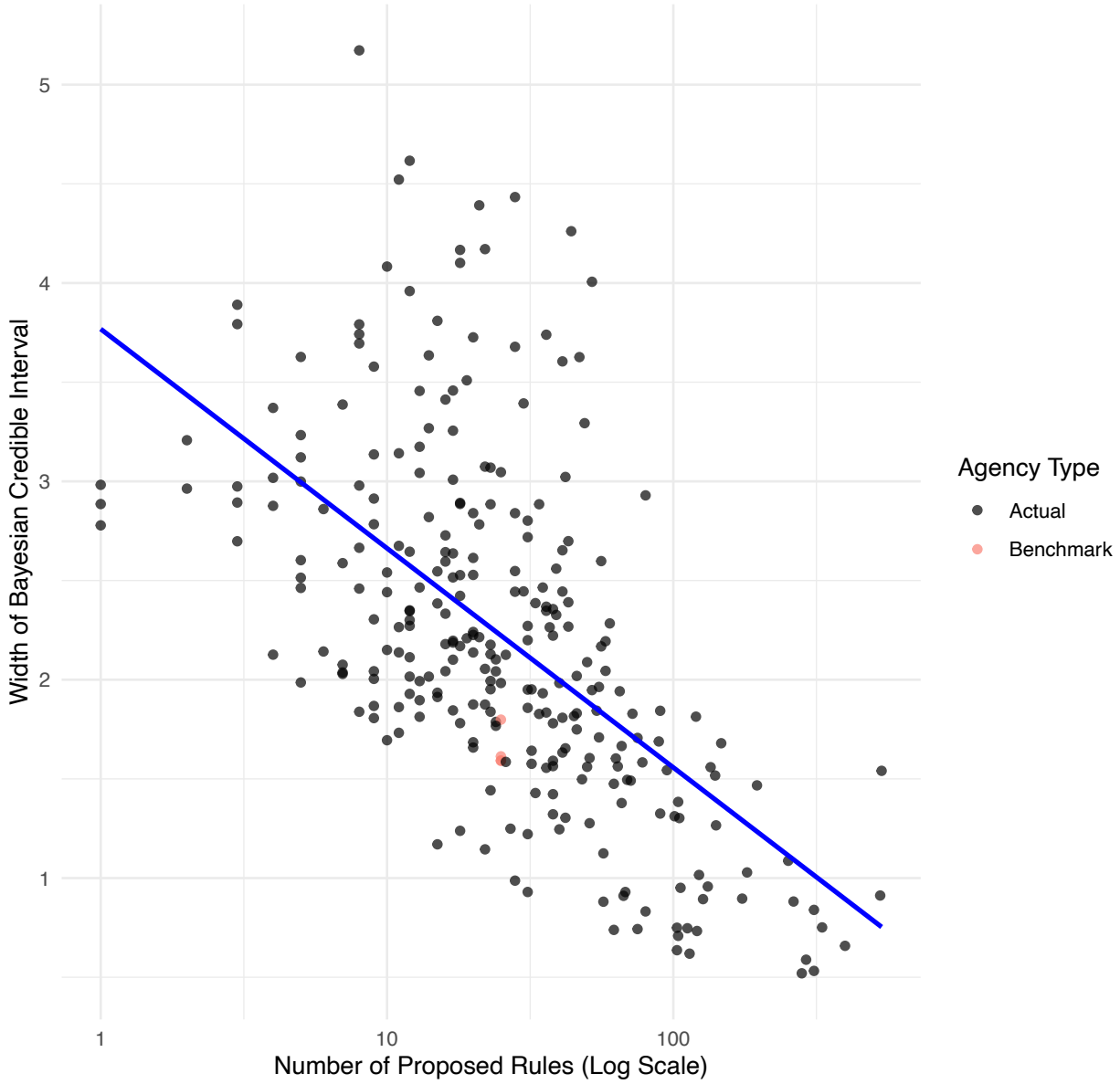
B Specification of Stan Model

```
data {
  // data dimensions: agencies (A), presidents (P), rule significance (S), observations (N), time periods (T)
  int<lower=1> A;
  int<lower=1> P;
  int<lower=1> S;
  int<lower=1> N;
  int<lower=1> T;
  // indices for each observation
  int<lower=1, upper=A> a[N];
  int<lower=1, upper=P> p[N];
  int<lower=1, upper=S> s[N];
  int<lower=0, upper=1> y[N];
  int<lower=1, upper=T> t[N];
}
parameters {
  vector[P] xpresident;
  matrix[T, A] xagency_raw;       // non-centered RW innovations
  real<lower=0> sigma_rw;         // RW step size
  real mu_kappa;                       // kappa mean
  real<lower=0> tau_kappa;        // kappa sd (heterogeneity)
  vector[A] kappa_raw;            // non-centered agency intercepts
  vector[S] gamma;
  vector<lower=0>[S] eta;
}
transformed parameters {
  matrix[T, A] xagency;   // dynamic agency positions
  vector[A] kappa;          // realized agency intercepts
  xagency[1] = xagency_raw[1];   // non-centered random walk
  for (i in 2:T)
    for (j in 1:A)
      xagency[i, j] = xagency[i - 1, j] + sigma_rw * xagency_raw[i, j];
  kappa = mu_kappa + tau_kappa * kappa_raw;   // kappa's hierarchical structure
}
model {
  to_vector(xagency_raw) ~ normal(0, 1);
  sigma_rw ~ normal(0, 1);
  // hierarchical priors on kappa
  mu_kappa  ~ normal(0, 1);
```

```
  tau_kappa ~ normal(0, 1);
  kappa_raw ~ normal(0, 1);
  gamma ~ normal(0, 1);
  eta  ~ lognormal(-2, 0.8);  // small, positive prior
  xpresident ~ normal(0, 1);
  xpresident[1] ~ normal(-0.438, 0.01);  // Clinton
  xpresident[2] ~ normal( 0.693, 0.01);  // Bush
  xpresident[3] ~ normal(-0.358, 0.01);  // Obama
  xpresident[4] ~ normal( 0.403, 0.01);  // Trump
  {   // probit likelihood with rescaled distance
    real c = 1.5;        // rescaling constant; puts distance, d, on the probit latent scale (sd = 1)
    vector[N] alpha;
    for (i in 1:N) {
      real d = (xpresident[p[i]] - xagency[t[i], a[i]]) / c;
      alpha[i] = kappa[a[i]] + gamma[s[i]] + eta[s[i]] * d * d;
    }
    y ~ bernoulli(Phi(alpha));               // vectorized Bernoulli with probit
  }
}
```

C Uncertainty in Ideal Point Estimates

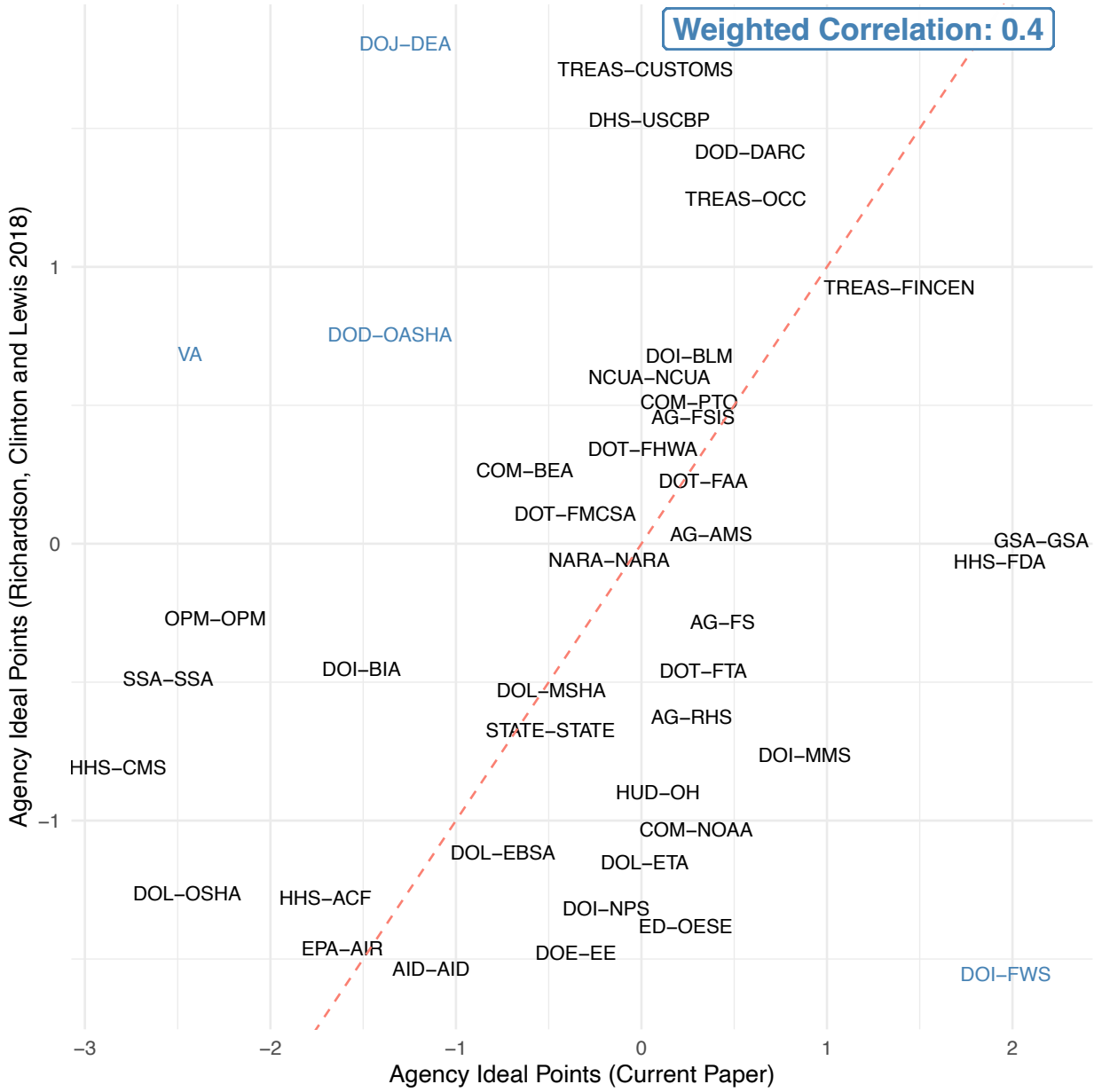
Figure A1: Precision of Ideal Point Esimates



Notes: Each dot represents one of the 280 agency ideal points, including the benchmark agency. As discussed, the benchmark agency's productivity, as measured by the number of proposed rules, is similar to that of the median agency.

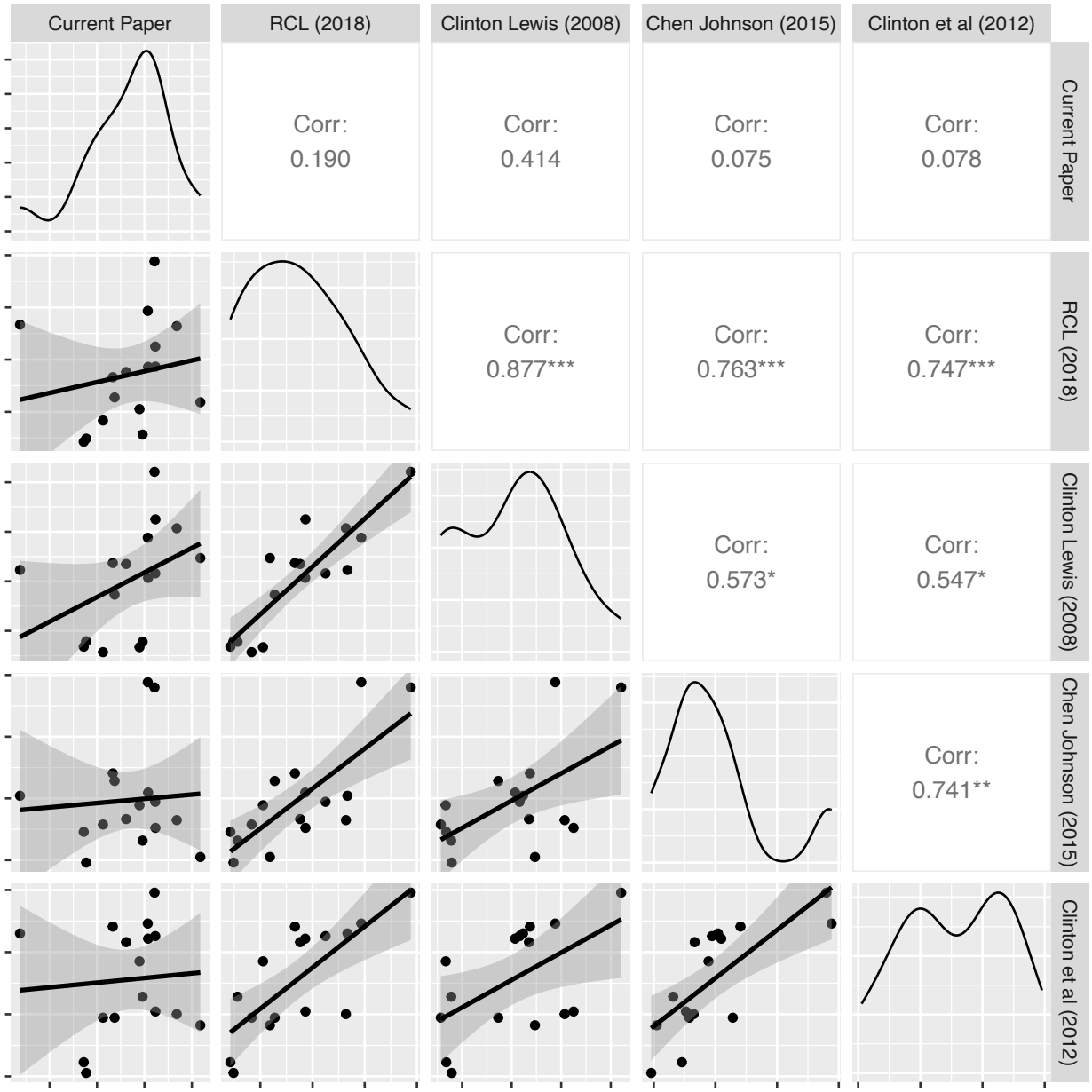
D Comparisons to Other Measures

Figure A2: Correlation with an Alternative Measure



Notes: Appendix A includes complete names for agencies and departments. The correlation excludes four outliers: VA, DOD-OASHA, DOI-FWS, DOJ-DEA. The dashed line is the 45-degree line.

Figure A3: Department-level Correlations with Other Measures



Notes: Each dot represents one of the 16 departments depicted in Figure 3 and also in Appendix Table A3.

Table A3: Ranking of Department Ideology (Liberal to Conservative)

	Current Paper	Richardson et al (2018)	Chen-Johnson (2015)	Clinton et al (2012)	Clinton-Lewis (2008)
1	Veterans Affairs	Health	Environment	Environment	Labor
2	Health	Environment	Interior	Interior	Housing
3	Environment	Education	Education	Education	Health
4	Labor	Labor	Health	Health	Education
5	Justice	Housing	Commerce	Commerce	Environment
6	State	Interior	Labor	Labor	State
7	Energy	State	Treasury	Treasury	Transportation
8	Housing	Justice	Energy	Energy	Agriculture
9	Education	Energy	Housing	Housing	Veterans Affairs
10	Homeland Security	Transportation	Agriculture	Agriculture	Energy
11	Transportation	Commerce	Veterans Affairs	Veterans Affairs	Justice
12	Defense	Agriculture	Transportation	Transportation	Interior
13	Agriculture	Treasury	State	State	Homeland Security
14	Commerce	Veterans Affairs	Justice	Justice	Treasury
15	Treasury	Homeland Security	Defense	Defense	Commerce
16	Interior	Defense	Homeland Security	Homeland Security	Defense

Notes: Ideal points from Chen and Johnson (2015) are averaged across administrations. Ideal points from Clinton et al. (2012) use the survey responses from career bureaucrats.

E External Validation of Trends in Agency Ideology

Metadata from Regulations.gov was collected for rules issued by each agency under study: the EPA's Office of Air, the Office of Personnel Management (OPM), and the Department of Veterans Affairs (VA). Because comment data from the Bush administration is often incomplete, the analysis focuses on the Obama and Trump administrations. This is also the period during which the agency ideal points exhibit the most pronounced conservative shifts, as discussed in the manuscript.

The ideological orientation of the selected stakeholder groups is generally well known, but it can also be inferred from established measurement models, such as those developed by Bonica (2013) and Crosson, Furnas, and Lorenz (2020). The Chamber of Commerce scores 0.88 in both measures (scores are standardized with a mean of 0 and a standard deviation of 1, where positive values indicate conservatism), while the AFGE scores -1.16 and -1.28, respectively. The AMA, more moderate by comparison, scores 0.14 and -0.05, respectively.

The Regulations.gov metadata for these three agencies includes sufficient detail about the organization submitting each comment. These records were used to identify proposed rules that received input from the key stakeholders mentioned above. For each identified comment, the full text was retrieved and coded based on whether the stakeholder expressed support, opposition, or neutrality toward the proposed rule. A comment was coded as neutral if it contained no clear statement of support or opposition for the rule's overall direction, but instead offered recommendations for revision in a technical or non-committal tone.

Among the three examined cases, the EPA's Office of Air provided the most complete metadata, with an organization listed for every comment. OPM's metadata was usable, though less complete. Although a metadata field for the organization name exists, it is frequently missing, absent in 449 of the 2,278 comments analyzed. Fortunately, each comment also includes a categorical designation of the commenter—such as citizen, agency, profes-

sional organization, or union—which facilitated further filtering. To identify comments from the American Federation of Government Employees (AFGE) and the National Federation of Federal Employees (NFFE)—two of the major unions representing federal workers—all comments that either listed these organizations by name or were categorized as "union" were downloaded and coded. The vast majority of union-designated comments lacking an explicit organization name were submitted by individual union members and not the leadership, and thus did not necessarily represent the union's official position. Additional comments from local branches were similarly excluded, as they also did not reflect the stance of the national organization.

The VA's metadata was the weakest of the three cases. Only one comment across all VA rulemakings explicitly listed the American Medical Association (AMA) in the *organization* field. To supplement this, a keyword search for "American Medical Association" was conducted on the list of commenters provided under each proposed rules on Regulations.gov, which returned three additional AMA comments. The results are shown in Table A4.

Table A4: Stakeholder Positions on Agency Rulemakings (Obama and Trump admins.)

Stakeholder (Agency in Parentheses)	Position	Obama	Trump
Chamber of Commerce (EPA's Office of Air)	Neutral	1	0
	Oppose	11	0
	Support	0	8
AFGE and AFFE (Office of Personnel Management)	Neutral	3	1
	Oppose	0	4
	Support	3	0
American Medical Association (Veterans' Affairs)	Neutral	0	0
	Oppose	1	0
	Support	0	3

References

Bonica, Adam. 2013. "Ideology and Interests in the Political Marketplace." *American Journal of Political Science* 57(2): 294–311.

Crosson, Jesse M., Alexander C. Furnas, and Geoffrey M. Lorenz. 2020. "Polarized Pluralism: Organizational Preferences and Biases in the American Pressure System." *American Political Science Review* 114(4): 1117–1137.